

Sam Acquaviva

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Education

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Master of Engineering, Computer Science and Cognitive Science

May 2024

- GPA: 5.0/5.0
- MEng Thesis: *Overcoming the Expressivity-Efficiency Tradeoff in Program Induction*. Advisors: Josh Tenenbaum, Yewen Pu

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Bachelor of Science, Computer Science and Cognitive Science

May 2023

- GPA: 4.9/5.0
- Relevant coursework: Advanced Natural Language Processing, Matrix Methods for Machine Learning, Deep Learning, Tissue vs. Silicon: Differences in Machine Learning, Algorithms, Signal Processing,

Work Experience

After Thought AI (venture-backed neo-lab)

San Francisco, CA

Founding Research Scientist

Sep 2024 – Present

- Working on novel architectures and training recipes for reasoning with code (details not public, reach out)

Apple

Seattle, WA

Machine Learning Intern

May 2022 – August 2022

- Designed and implemented semi-supervised training pipeline for Siri from high-risk production data, catching hundreds of surprising failure modes that led to real production fixes

X, The Moonshot Factory (Google X)

Mountain View, CA

AI Residency

Feb 2022 – May 2022

- Extended state-of-the-art neural-guided program synthesis engine (DreamCoder) to new reasoning-heavy domains

John Deere

Remote

Machine Learning Intern

Jun 2021 – Sep 2021

- Developed computer vision system to identify and count corn kernels from noisy production images
- Automated advanced skid steer maneuvers in simulation using reinforcement learning (PPO)

Publications

- **S. Acquaviva**, Y. Pu, M. Kryven, T. Sechopoulos, C. Wong, G. Ecanow, M. Nye, M. Tessler, J. Tenenbaum. *Communicating Natural Programs to Humans and Machines*. NeurIPS 2022, Datasets & Benchmarks Track.
- G. Leverick, S. Feng, P. Acosta, **S. Acquaviva**, F. Bardé, S. Cotte, Y. Shao-Horn. *Tunable Redox Mediators for Li-O₂ Batteries Based on Interhalide Complexes*. ACS Applied Materials & Interfaces, 2022.
- G. Ecanow, C. Wong, **S. Acquaviva**, Y. Pu, M. Kryven, J. Tenenbaum. *Core knowledge objects in reasoning and language use for highly abstract inductive tasks*. CogSci 2021.

Projects and Activities

LLM Pretraining records

Feb - Apr 2026

- Set many records on competitive LLM pretraining competitions: modded-nanogpt, Q Labs slowrun, OpenAI parameter golf

FlashOptim in JAX

March 2026

- Wrote custom kernels and quantization tricks to halve the memory consumption of optimizers like Adam or SGD in JAX

MIT Activities

Sep 2019 - May 2024

- MIT Class of 2023 Vice President (2019-2020)
- Varsity track/cross country: led team to MIT's first NCAA team championship, in any sport
- US Track/Cross Country Association Men's Scholar-Athlete of the Year (2022)

Additional

Technical Skills: JAX, PyTorch, Pallas, Triton, Python, human-in-the-loop data pipelines, RL for LLMs, systems optimization